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ORIGINAL ARTICLE

Process optimization of material extrusion additive manufacturing with ASA: robust design and predictive models for engineering response metrics

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Abstract

Acrylonitrile styrene acrylate (ASA) is increasingly popular among traditional engineering polymers for material extrusion (MEX) additive manufacturing (AM). Satisfactory printability and engineering metrics and excellent stability in hard environmental conditions explain this status. To achieve the highest possible performance levels, the optimization of the generic printing control settings is critical. This paper focuses on delivering enhanced mechanical and morphological characteristics when printing with ASA. To meet these goals, an L25 orthogonal robust design was compiled and executed, enabling the simultaneous assessment of six (6) control parameters, i.e., the strand width (S_W), infill orientation (angle) (A_{INF}), layer thickness (T_L), printing speed (P_S), nozzle temperature (T_N), and bed temperature (T_B). A massive experimental course with two hundred and ninety specimens was engaged to deliver response metrics, such as tensile ultimate strength (σ_B^T), tensile yield strength (σ_B^Y), tensile modulus of elasticity (E^T), flexural strength (σ_B^F), and flexural modulus of elasticity (E^F). Three modeling methods were applied to compare their efficacy in these loading scenarios: the Reduced Quadratic Regression Model (RQRM), Linear Regression Model (LRM), and Quadratic Regression Model (QRM). RQRM and QRM achieved similar prediction accuracies, whereas LRM yielded poorer predictive results than the other methods. Confirmation runs validated the validity of the compiled prediction functions, thereby making them suitable for direct industrial use. The most influential parameter was the A_{INF}. The optimized 3D printing settings improved all mechanical performance metrics by more than 40%, while the flexural strength was improved by more than 60%.

Keyword Acrylonitrile-styrene-acrylate (ASA); Orthogonal design; Three-dimensional (3D) printing control parameters; Morphological evaluation

1 Introduction

Fused filament fabrication (FFF) [1] is a well-known technology that belongs to the general category of additive manufacturing (AM) and contributes to service applications in several fields [2]. AM can be applied to automotive [3–6], aerospace [7–9], medicine [10–13], dentistry [14–16], electronics, and so on [17–24]. Advantages such as speed, sustainability, and flexibility are some of the many [25, 26] that contribute to the use of three-dimensional (3D) printing.

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It should be highlighted that the material resources, and consequently their waste, are limited as the products are manufactured layer-by-layer without having excess material, as happens in the case of cutting the desired part from a volume [27]. 3D-printed objects are created, each of which has a unique structure [28, 29] and different properties [30]. There is a wide range of possible combinations of mechanical properties that can determine the resulting behavior and mechanical properties of manufactured products [31].

Consequently, it is important to conduct thorough research on the properties of the materials and the interaction of the chosen 3D printing settings to achieve the optimum results [32, 33]. To manage the market and industry demands, in combination with environmental and economic events [34, 35], there is a need to develop polymeric materials and composites with superior properties and capabilities [36]. Commonly used polymeric materials used in the method are Polylactic Acid (PLA) [37], Acrylonitrile Butadiene Styrene (ABS) [38], polyamides (PA), such as PA12 [29], high-density polyethylene (HDPE) [39], polycarbonate (PC) [40], Thermoplastic Polyurethane (TPU) [41], and polypropylene (PP) [42, 43], among other, in pure or composites (micro or nano) form, as the bibliography investigation revealed. The variety of polymers is attributed to their different properties, making them suitable for specific types of applications. For example, TPU has a rubber-like response, while PC is tough and brittle.

Recently, 3D printing technology has employed acrylonitrile-styrene-acrylate (ASA) [44], which is a promising material for operation under challenging conditions. It is characterized by a morphology [45] and characteristics [46] similar to those of acrylonitrile–butadiene–styrene (ABS). However, the ability of ASA to withstand harsh conditions [47] makes it a better option than ABS in their respective applications [48]. ASA, like many thermoplastics, can be tricky to print in some cases. ASA performs well in demanding conditions, but during printing, it can still show problems such as warping or moisture absorption. These can affect how layers stick together or the final surface quality, so careful adjustment of the printing settings is necessary to get reliable results. These are additional aspects that strengthen the need for optimization.

Characteristics such as the mechanical, thermal, and functional behavior of ASA materials are the reason why they are preferred for being applied to a wide variety of applications, such as outdoor applications [49] and automotive [50] applications. It possesses weather resistance, UV stability [51], and layer-to-layer adhesion [52, 53]. Chemical resistance is an important characteristic of ASA [54]. UV resistance [55] can prevent the degradation of material properties under common outdoor conditions [56].

With regard to the ASA market, several reports have cited information derived from their predictions, each on various forecast periods. Mordor Intelligence reported [57] an ASA filament 3D printing material market size for 2024–2029, where a 21.87% CAGR was estimated starting from USD 156.32 million in 2024 and resulting in USD 420.27 million by 2029. In addition, there was another report from Mordor Intelligence examining the forecast period of 2025–2030 and suggested a CAGR of 21.87%, from USD 190.49 million (2025) to USD 512.12 million (2030) [58].

Knowledge-sourcing intelligence [59] stated that from USD 59.179 million in 2020, USD 259.040 million would be reached by 2027, with a 23.48% CAGR. The market size of Global 3D Printer ASA Filament in 2023 was projected to be USD 0.33 billion and then expected to reach USD 1.94 billion by the year 2032 (21.6% CAGR), according to another report[60]. An additional report [61] stated that between 2020 and 2027, there would be a 23.5% CAGR increase from USD 59.18 million to USD 259.04 million.

Several researchers have conducted optimization investigations [62–66] in an effort to achieve the mechanical characterization of various response parameters and how they are affected by the printing parameters [67, 68]. To this end, there are a few different supporting tools specializing in conducting such experimental procedures more easily and with reliability. A few of them are the full factorial design method [69, 70], the Behnken design [71, 72], and Taguchi's design [32, 73] of experiments.

The Taguchi DOE (Design of experiments—selected herein) is a tool used for the preparation of an experiment with the assistance of the selected independent process variables [74]. The Taguchi method has been employed for the optimization of samples [75, 76] produced from various materials, including PLA [69, 71, 73, 77–83], (PC) [84], polyamides (PA12 or PA6) [32, 70, 72, 85, 86], PP [87], PMMA [88], and ASA [75, 76, 89]. It has been applied efficiently in high-performance polymers, such as PEEK [90–94], PEKK [95], and (PEI) [96], too, or in biocompatible polymers in bioplotting, such as PCL [97]. Such methods have been applied to optimize the quality metrics of the 3D printed parts, apart from their mechanical properties [98]. Research has demonstrated that when employing optimized parameter configurations, the strength can be enhanced by up to 300% compared to producing components with a standard set of parameters [96]. Consequently, these modeling techniques are widely recognized as the standard approach in the field for optimizing process parameters, specifically in 3D printing with the material extrusion (MEX) technique.

Attempts have been made to optimize the 3D printing settings of ASA parts, such as the study of Magri et al. [51], in which the response surface methodology was applied to three 3D printing settings to locate the best parameter values regarding tensile strength. In the study of Hameed et al. [99], ASA material was employed in an attempt to optimize its 3D printing parameters, with the aim of achieving enhanced

mechanical properties by utilizing the L18 orthogonal array Taguchi experimental design. Five process parameters were chosen, of which the infill density and layer height presented the greatest impact. In the study of Kumar et al. [44], two 3D printing parameters were assessed (infill percentage and build orientation) for their effect on tensile and flexural loading scenarios. In the study of Rakshit et al. [47], ASA specimens were produced according to the chosen design of experiment (DOE), with layer thickness, orientation angle, and infill density as input parameters and ultimate tensile strength as the response variable.

Moreover, the impact of manufacturing parameters on the mechanical and tribological behavior of ASA was investigated in this [100] study, considering two printing parameters and proving that the layer thickness has an influence on them. ASA has also been part of the composites investigated in the literature, combined with bamboo fiber and elastomer-chlorinated polyethylene [101], titanium dioxide [102], polyetheretherketone [103], poly(vinyl chloride) [104], carbon fiber [105–107], glass fiber, polycarbonate [108], Polybutylene Terephthalate [109], walnut shell-filled [110], and antimony trioxide [111].

The literature review revealed the academic and industry interest in the ASA polymer. Still, in MEX 3D printing literature is nowadays rather narrow. Studies have been presented for the optimization of the 3D settings of ASA items, considering only two parameters [44], or up to five parameters [99], for parts tested under tensile loading. Herein, a massive effort with six 3D printing settings, two types of mechanical testing (tensile and flexural), and three modeling methods, was carried out.

In the realm of experimental design, the incorporation of additional parameters within a Taguchi design framework offers distinct methodological advantages that enhance both the depth and breadth of analysis. A primary benefit is the attainment of greater insight into the investigative system. By examining a broader array of factors simultaneously, researchers are more likely to identify variables that significantly impact performance measures, a critical consideration when numerous interdependent factors are present. Furthermore, the inclusion of a larger number of parameters enhances the potential to discern interaction and synergistic effects, which might otherwise remain obscured in more conservative designs. Furthermore, herein, a much deeper and broader analysis was carried out, with the application of three regression models, the comparison between them, the formation of prediction models, and their validation with additional confirmation runs, as mentioned.

Analysis of the effect of six 3D printing parameters on Acrylonitrile Styrene Acrylate (ASA) material using a combination of Taguchi design, Analysis of Variance (ANOVA), and regression modeling offers an integrated and methodologically robust framework for process optimization and performance study. ASA, with its superior mechanical stability, weather resistance, and UV durability, has wide usage in functional and outdoor additive manufacturing applications. However, 3D-printed ASA part behavior is largely governed by the selection and interaction of printing parameters. A parameter-controlled study with these six parameters provides a robust evaluation of the individual and interactive effects of these parameters on key performance measures such as tensile strength, dimensional stability, and surface finish. Taguchi's design use provides an economical experimental framework through orthogonal arrays such that parameters can be examined with a limited number of experimental runs. The method reduces resource usage and experimental complexity without affecting the ability to detect the important effects of each parameter. ANOVA fills in the gaps by offering a statistically sound method for quantifying the relative contribution of each factor, distinguishing between which parameters contribute statistically significantly to the desired outcome. This is particularly important in additive manufacturing, where minute changes in parameters can make dramatic differences in part quality. Furthermore, regression modeling is a predictive technique by which empirical response variable-input parameter relationships can be determined. By developing mathematical models from experimental data, it is possible to forecast performance results for various combinations of parameters, thus enabling informed decision-making and process control. The integration of Taguchi design, ANOVA, and regression analysis not only facilitates the easy identification of optimal printing conditions for ASA but also enhances the reproducibility and robustness of the process. The three-pronged approach collectively contributes significantly to the parametric sensitivity of ASA in fused filament fabrication and encourages the development of data-driven optimization approaches for performance optimization in industry and research communities alike.

A total of two hundred and ninety samples were assessed. Optimization of the experimental samples was attempted by examining the influence of strand width (S_W), infill angle (A_{INF}), layer thickness (T_L), printing speed (P_S), nozzle temperature (T_N), and bed temperature (T_B) control parameters, on the tensile ultimate strength (σ_B^T), tensile yield strength (σ_B^Y), tensile modulus of elasticity (E^T), flexural strength (σ_B^F) and flexural modulus of elasticity (E^F) response outputs. The experimental design selected for this study was the Taguchi orthogonal L25. Furthermore, the experimental data were assessed with three different approaches. More specifically, the Reduced Quadratic Regression Model (RQRM), Linear Regression Model (LRM), and Quadratic Regression Model (QRM) were applied to benchmark their effectiveness. Models for the prediction of the output metrics were compiled and correlated with findings from confirmation runs, to

prove their usefulness for real-life applications. For completeness, apart from the mechanical behavior, thermal characteristics, morphology, and structure of the samples were investigated. According to the bibliographic research conducted, this is the first time such a thorough and wide analysis is presented for the optimization of the 3D printing settings of ASA examples made with the MEX method. The findings can be exploited as a road map for selecting proper settings when building parts with the ASA polymer, leading to more robust parts, which are in high demand in applications.

2 Materials and methods

2.1 Material

The ASA material supplier chosen for this investigation was eSUN, located in Microsoft Comtech Tower (Shenzhen, China). According to the supplier's information, the purchased material is weather-resistant, while it is also resistant to high impact. Additionally, it possesses high toughness, rigidity, density of 1 g/cm^3, and melt flow index of 10–15 (220 °C/10 kg).

2.2 Experimental procedure

The steps included in the experimental modeling strategy selected herein are presented in Fig. 1, along with images captured during some of the procedures.

2.3 Thermal examination of ASA material

Thermal characterization of the ASA material was conducted to investigate the effect of temperature on its degradation. Two analyses were performed: thermogravimetric analysis (TGA) and differential scanning calorimetry (DSC). The devices employed were a Perkin Elmer Diamond (Waltham, Massachusetts, USA) and a TA Instruments Discovery-Series DSC 25 (Delaware, USA). Figure 2 shows the curves derived from the TGA (Fig. 2a) and DSC (Fig. 2b) results, as well as the measured temperatures. Measurements were taken in a nitrogen atmosphere, following typical settings from the literature [36].

2.4 3D Specimen manufacturing through MEX

The tensile and flexural specimen production was implemented with the MEX AM 3D printing technology. The corresponding necessary information provided by ASTM

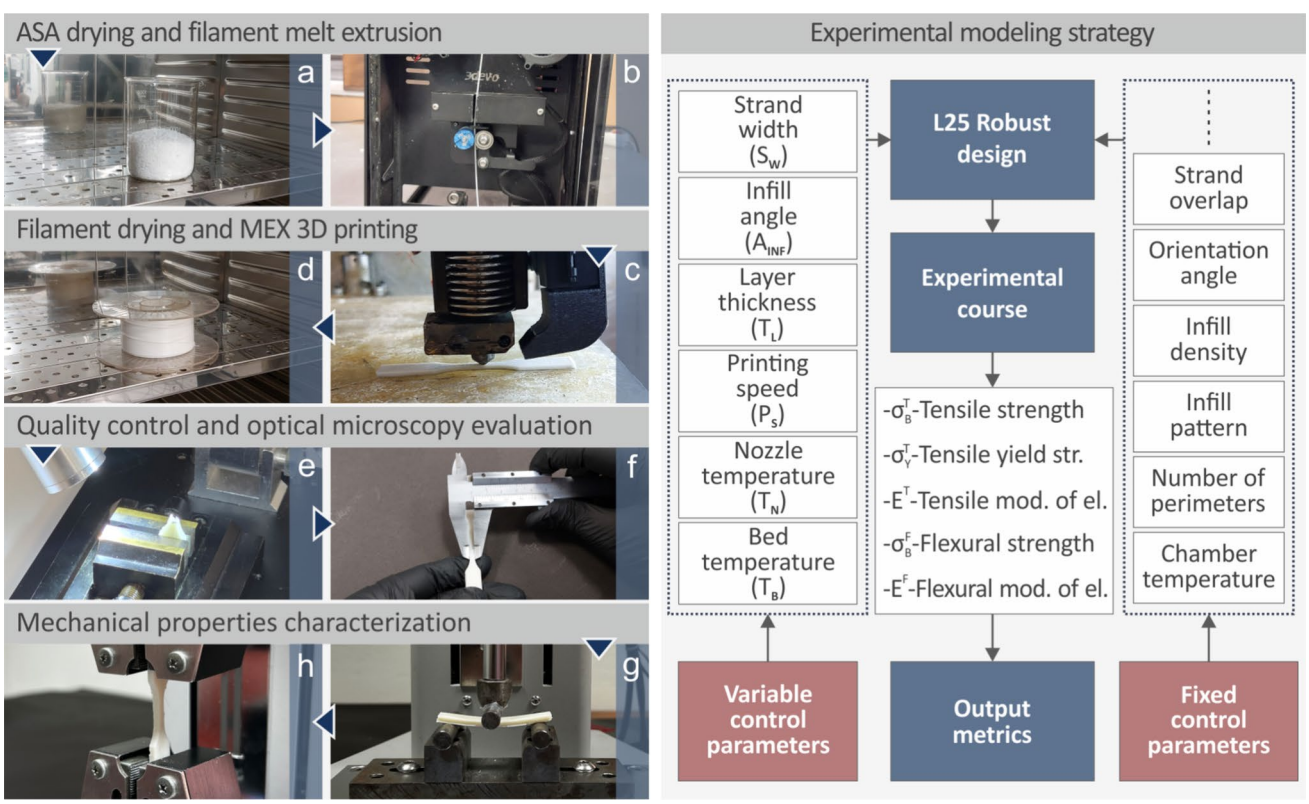


Fig. 1 Images captured during the different procedures conducted for this investigation and experimental modeling strategy presentation in a diagram

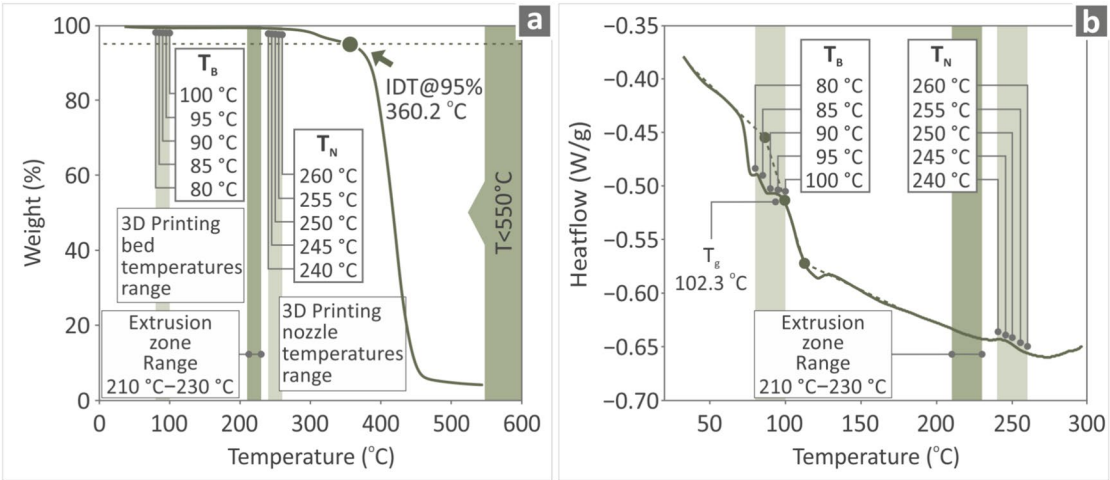
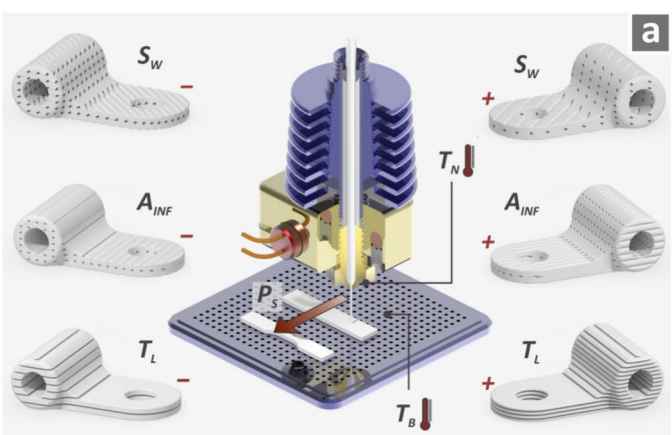


Fig. 2 (a) TGA and (b) DSC curves indicating the thermal behavior of the ASA samples

D790 (flexural tests) and ASTM D638 (tensile tests) international standards were followed and taken into consideration. Figure 3a shows the specimen manufacturing process in an illustration, and Fig. 3b lists the ASA printing parameter values considered in this study. Figure 3c shows the tensile and flexural specimens, their dimensions, and the standards followed. Before mechanical testing, all printed specimens were inspected and lightly cleaned to remove any surface irregularities, such as elephant foot or excess material at the base, ensuring proper contact and accurate results during the tests. Still, the surface quality is one of the characteristics of the 3D printing structures and the samples were tested with this feature. The surface quality was a typical one for 3D printing parts. The effect it has on mechanical performance is considered in the experimental results in this way, as the aim was to measure their strength.

2.5 Morphological evaluation

A specimen morphology investigation was conducted to observe their morphological characteristics belonging to their lateral surfaces, as well as their fractured surfaces



ASA Printing Parameters	Value	Units
Strand width	50–75–100–125–150	%
Infill angle	0.0–22.5–45.0–67.5–90.0	deg
Layer thickness	0.10–0.15–0.20–0.25–0.30	mm
Printing speed	20–30–40–50–60	mm/s
Nozzle temperature	240–245–250–255–260	°C
Bed temperature	80–85–90–95–100	°C
Orientation angle	0	deg
Infill density	100	%
Infill pattern	Linear	–
Number of perimeters	2	–

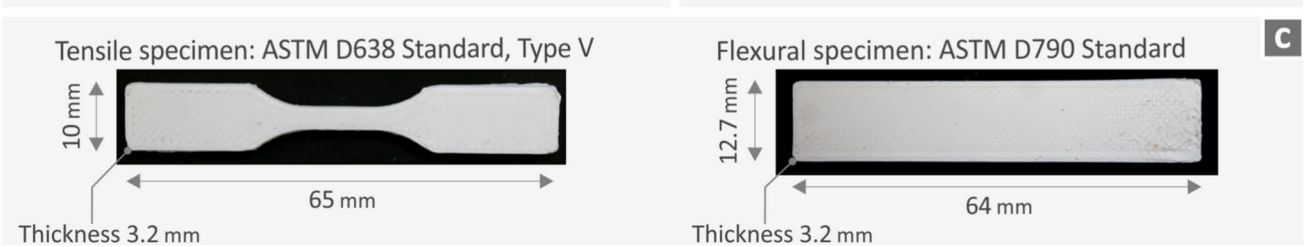


Fig. 3 (a) Illustration of specimen manufacturing, (b) ASA printing parameters, (c) specimen design, dimensions and followed standards

(tensile samples only), after testing. The Scanning Electron Microscopy (SEM) analysis was used for this purpose and carried out by a JSM 6362LV (Jeol Ltd., Peabody, Massachusetts, USA, Au sputtering on samples, high-vacuum, 20 kV). A microscopic examination of the samples was performed using a digital microscope model named Levenhuk DTX RC4, featuring a 7-inch rotatable LCD screen and a built-in 4MP CMOS camera, manufactured by Levenhuk, Inc. (Tampa, FL, USA), allowing real-time observation and image capture via a connected computer.

2.6 Taguchi DOE, ANOVA, and regression methods

The Taguchi method was first used to select cases for modeling from an orthogonal array. Classical experimental design methods are complex and thus not easy to use. Additionally, a large number of experiments or simulations must be performed when the number of process parameters increases.

The Taguchi approach employs a distinctive arrangement of orthogonal arrays to examine the complete parameter space, enabling the resolution of problems with significantly fewer experiments [112]. An orthogonal array is typically chosen based on the overall degree of freedom (DOF). The method is presented analytically and documented with literature [113–123] in the supplementary information.

Hereby, Taguchi L25 orthogonal array was selected to be used, and twenty-five (25) experiments were conducted, five (5) repetitions each, resulting in one hundred and twenty-five (125) response sets. Six (6) control parameters were chosen, namely S_W, A_{INF}, T_L, P_S, T_N, and T_B, as listed in Table 1 along with their levels, which were chosen while taking into consideration the related literature on modeling and optimization of the MEX 3D printing setting values [32, 70, 72, 82–84, 88] and preliminary screening tests.

ANOVA followed for the experimental outcome interpretation by evaluating each parameter's contribution ratio [124]. The method is analytically presented along with the equations applied in the supplementary information. Furthermore, three regression models were applied, i.e., Reduced Quadratic Regression Model (RQRM), Linear Regression Model (LRM), and Quadratic Regression Model (QRM), The respective equations of each method are provided in the supplementary information. These were applied to form a prediction model for every output metric as an equation of the control parameters. Each regression method produced respective equations from simpler (LRM, first-degree equations) to more advanced ones (RQRM and

Table 1 Taguchi L25 design of experiments: examined control parameters and levels

Run	$S_W(\%)$	$A_{INF}(deg)$	$T_L(mm)$	$P_S(mm/sec)$	$T_N(°C)$	$T_B(°C)$
1	50	0.0	0.10	20	240	80
2	50	22.5	0.15	30	245	85
3	50	45.0	0.20	40	250	90
4	50	67.5	0.25	50	255	95
5	50	90.0	0.30	60	260	100
6	75	0.0	0.15	40	255	100
7	75	22.5	0.20	50	260	80
8	75	45.0	0.25	60	240	85
9	75	67.5	0.30	20	245	90
10	75	90.0	0.10	30	250	95
11	100	0.0	0.20	60	245	95
12	100	22.5	0.25	20	250	100
13	100	45.0	0.30	30	255	80
14	100	67.5	0.10	40	260	85
15	100	90.0	0.15	50	240	90
16	125	0.0	0.25	30	260	90
17	125	22.5	0.30	40	240	95
18	125	45.0	0.10	50	245	100
19	125	67.5	0.15	60	250	80
20	125	90.0	0.20	20	255	85
21	150	0.0	0.30	50	250	85
22	150	22.5	0.10	60	255	90
23	150	45.0	0.15	20	260	95
24	150	67.5	0.20	30	240	100
25	150	90.0	0.25	40	245	80

QRM). This aimed to compare the accuracy of each method and determine if simpler equations in this case are sufficient for its modeling. Two additional confirmation runs were conducted to further assess the predictive models' precision, in which the experimental findings were correlated with the predicted values for the response metrics.

3 Results

3.1 Tensile and flexural derived experimental data

Figure 4 and Fig. 5 show the tensile and flexural results, respectively (runs 1, 7, 13, 19, and 25), for the specimens from runs of the created diagonal. Figure 4a–e presents the tensile stress–strain curves and the printing parameters of runs of the formed L25 array. Figure 4f–j depicts the respective SEM images (27 × magnification) from the fractured side of a specimen chosen to represent each run. It can be observed that the samples from Runs 7 and 19 exhibit brittle behavior, whereas the rest of the samples are more ductile and present voids and defects. In run 1 the fracture surface collapsed, but still, the part failed to develop a higher strain than the remaining parts presented in the figure. It also developed higher ultimate strength compared to the remaining parts presented in the figure. The lowest strength and strain among the parts presented in the figure were developed in the part from run 25.

Figure 5 shows the corresponding flexural results for the samples derived from the same runs. Figure 5a–e illustrates the flexural, stress–strain curves, and 3D printing parameters of the runs from the diagonal of the array. Figure 5f–j shows the respective SEM lateral surface images (27 × magnification) of selected samples from each of the five runs, where it is revealed that the layering is characterized by good distribution, and no voids or discontinuities are detected, in all cases. In this test, the part from run 7 had the highest strength among the parts presented in the figure and again run 25 was the lowest one. The strain cannot be compared, as experiments were terminated at 5% strain, following the respective standard's instructions.

Figure 6 presents optical microscopy images (tensile samples, fracture surface) captured from all of the twenty-five (25) runs, along with their respective 3D printing settings, namely the S_W, A_{INF}, T_L, P_S, T_N, T_B. The optical microscopy images revealed brittle characteristics for runs 2, 6, 10, and 16, while the rest exhibited ductile behavior.

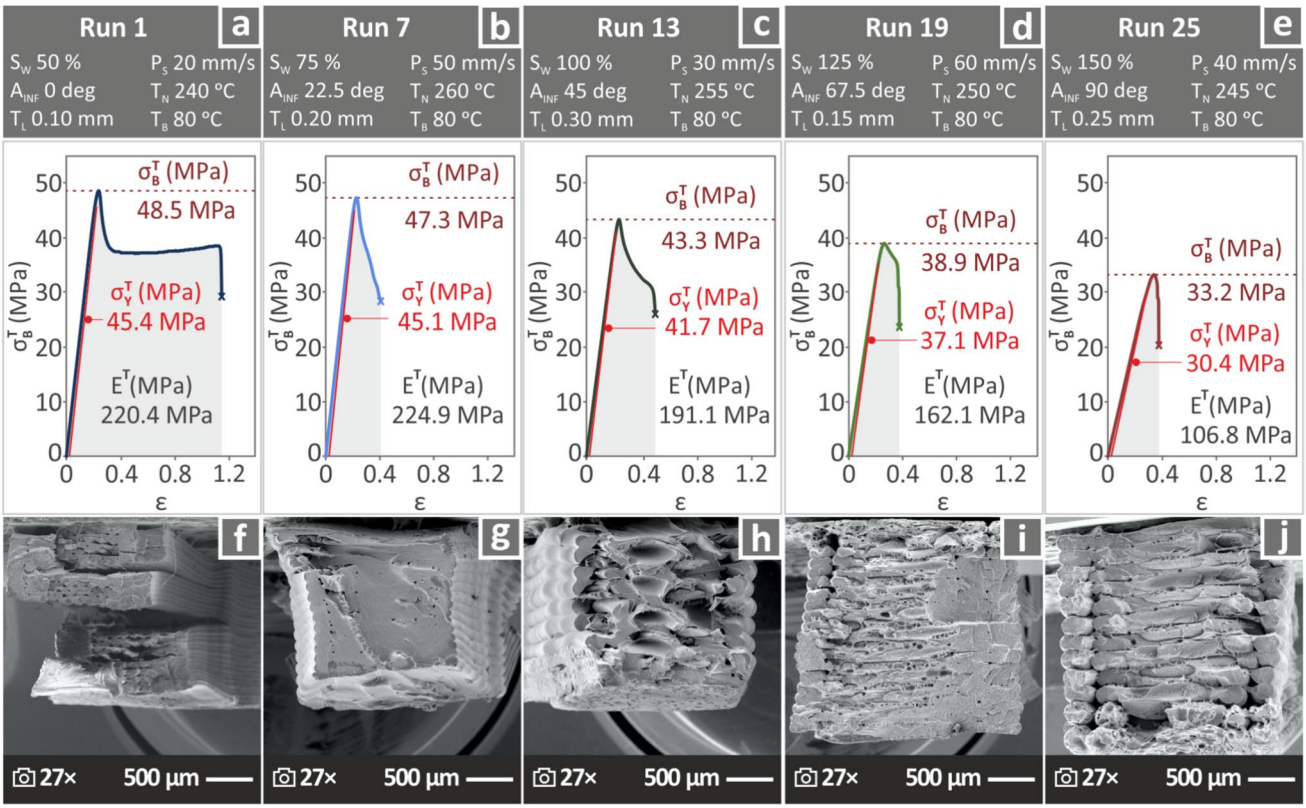


Fig. 4 (a–e) Tensile stress to strain curves of runs 1, 7, 13, 19, and 25 (runs from the diagonal) and their 3D printing settings, (f–j) SEM images of their fractured surfaces in 27 × magnification

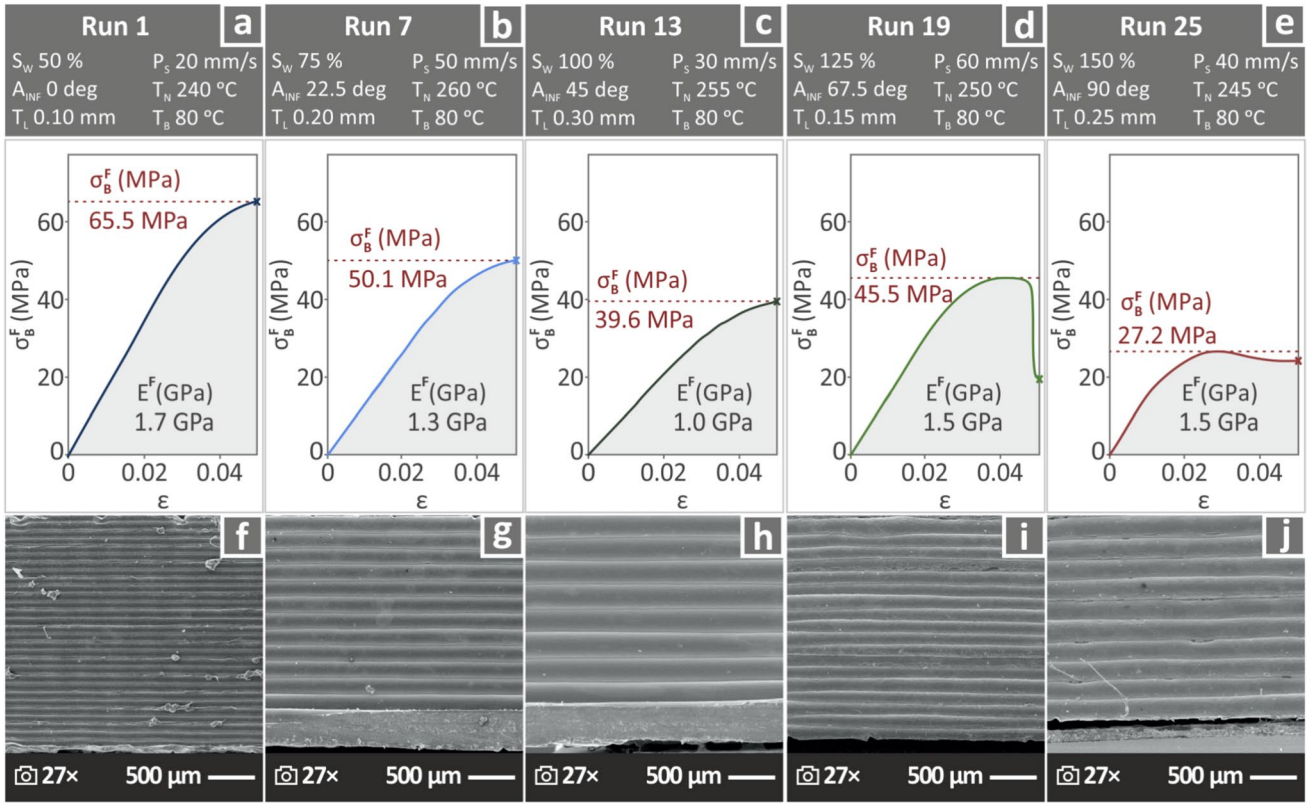


Fig. 5 (a–e) Flexural stress to strain curves of runs 1, 7, 13, 19, and 25 (runs from the diagonal) and their 3D printing settings, (f–j) SEM images of their lateral surfaces in 27 × magnification

3.2 Derived experimental data

In Table 2, these are the rankings for all of the control parameters S_W, A_{INF}, T_L, P_S, T_N, and T_B for means σ_B^T, σ_B^Y, E^T, σ_B^F and E^F, resulting after taking into consideration the delta statistics. For each factor, the difference measured by subtracting the lowest from the highest average value was calculated. In rank 1, the highest delta value had a significant effect on the value of the response parameter. In contrast, rank 6 had the lowest detected delta value, characterized by a low impact on the response value. According to this table, the raster angle was the highest-ranked parameter for all response metrics, except for the flexural modulus of elasticity. The lowest-ranked parameter differed among the response metrics.

Five (5) levels for each of the six (6) control parameters (S_W, A_{INF}, T_L, P_S, T_N, and T_B,) are listed in Table 2 along with the average values. In Table 3, these are the average and standard deviation values for all of the responses for σ_B^T, σ_B^Y, E^T, σ_B^F and E^F, for the 25 runs. In Table S1 found in the supplementary information, more derived data with regard to the 25 experimental runs are available.

3.3 ANOVA statistical technique derived data and regression model

The experimental results were processed with the assistance of the RQRM. The polynomial ANOVA tables show the RQRM values for each of the σ_B^T, σ_B^Y, E^T, σ_B^F and E^F versus the S_W, A_{INF}, T_L, P_S, T_N, and T_B are available in the supplementary information. In addition, below each table, the corresponding equation is utilized for each of the investigated response metrics (Eqs. (1)–(5)).

$$\sigma_B^T = 1863 + 0.0053 \times S_W + 0.0342 \times A_{INF} - 78.2 \times T_L + 0.3631 \times P_S - 14.42 \times T_N - 0.446 \times T_B - 0.000313 \times S_W^2 - 0.001482 \times A_{INF}^2 + 181.0 \times T_L^2 - 0.003693 \times P_S^2 + 0.02884 \times T_N^2 + 0.00331 \times T_B^2 \quad (1)$$

$$\sigma_Y^T = 1351 + 0.0126 \times S_W + 0.0707 \times A_{INF} - 78.6 \times T_L + 0.3209 \times P_S - 10.41 \times T_N - 0.346 \times T_B - 0.000372 \times S_W^2 - 0.001717 \times A_{INF}^2 + 183.7 \times T_L^2 - 0.003235 \times P_S^2 + 0.02090 \times T_N^2 + 0.00266 \times T_B^2 \quad (2)$$

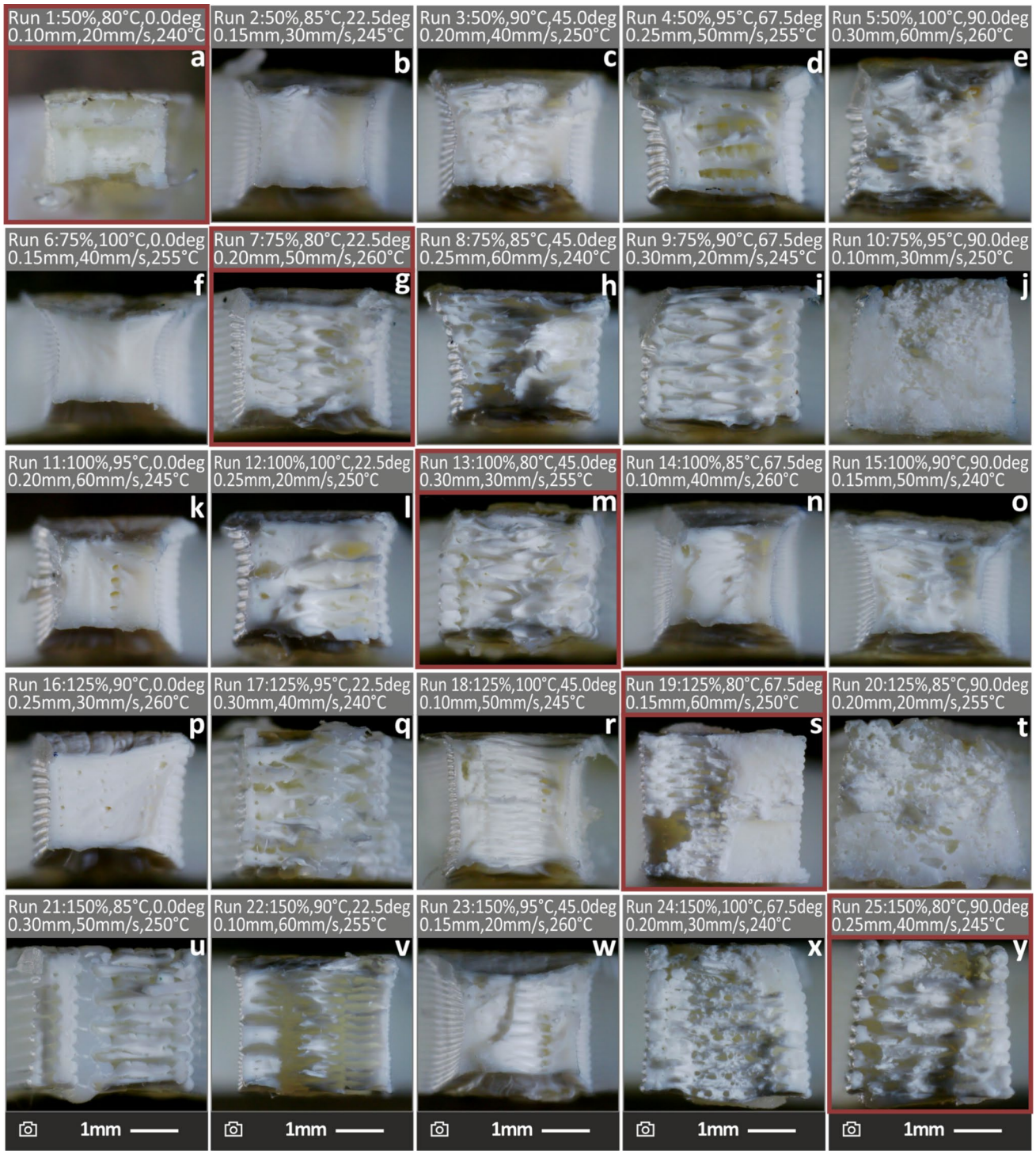


Fig. 6 Optical microscopy images of samples derived from the 25 runs and their respective 3D printing parameters

$$E^T = 8930 - 0.595 \times S_W + 0.046 \times A_{INF} - 398 \times T_L - 0.016 \times P_S - 81.1 \times T_N + 31.62 \times T_B + 0.00044 \times S_W^2 - 0.00822 \times A_{INF}^2 + 1019 \times T_L^2 + 0.00880 \times P_S^2 + 0.1632 \times T_N^2 - 0.1745 \times T_B^2 \quad (3)$$

$$\sigma_B^F = 4093 + 0.1601 \times S_W - 0.0998 \times A_{INF} - 392.0 \times T_L - 0.599 \times P_S - 39.87 \times T_N + 22.67 \times T_B - 0.000999 \times S_W^2 - 0.000850 \times A_{INF}^2 + 861.1 \times T_L^2 + 0.00981 \times P_S^2 + 0.07934 \times T_N^2 - 0.12597 \times T_B^2 \quad (4)$$

Table 2 Control parameters ranking for means σ_B^T, σ_B^Y, E^T, σ_B^F and E^F

Level	$S_W(\%)$	$A_{INF}(deg)$	$T_L(mm)$	$P_S(mm/sec)$	$T_N(°C)$	$T_B(°C)$
$\sigma_B^T(MPa)$						
1	45.78	46.63	44.73	41.44	44.59	42.3
2	44.92	45.69	43.61	42.6	43.06	41.84
3	44.21	44.93	41.83	44.39	41.86	43.62
4	41.57	42.41	43.02	44.45	42.27	44.14
5	40.3	37.11	43.58	43.89	44.99	44.88
Delta	5.48	9.52	2.9	3.02	3.12	3.04
Rank	2	1	6	5	3	4
$\sigma_Y^T(MPa)$						
1	43.65	43.44	42.44	39.41	41.56	40.12
2	42.93	43.3	41.38	40.41	40.79	39.82
3	42.05	42.91	39.51	41.92	40.05	41.51
4	39.16	40.62	40.88	42.28	40.44	41.6
5	37.82	35.34	41.41	41.59	42.77	42.56
Delta	5.83	8.09	2.93	2.87	2.71	2.74
Rank	2	1	3	4	6	5
$E^T(MPa)$						
1	216.8	218.5	196	181.9	196.8	184.6
2	209.5	208.3	194.5	184.3	185.2	191.4
3	196.2	202.1	184.2	191.7	187.3	205.6
4	172	186	192.4	200.3	190.4	199.2
5	172	151.6	199.5	208.3	206.9	185.7
Delta	44.8	66.9	15.2	26.4	21.7	20.9
Rank	2	1	6	3	4	5
$\sigma_B^F(MPa)$						
1	53.45	59.82	61.37	51.29	58.2	45.71
2	53.42	56.83	53.28	48.47	52.18	56.95
3	54.49	52	47.59	50.77	48.37	58.36
4	51.82	50.82	48.2	54.23	49.05	54.91
5	49.29	42.99	52.03	57.7	54.66	46.54
Delta	5.19	16.83	13.79	9.23	9.84	12.65
Rank	6	1	2	5	4	3
$E^F(MPa)$						
1	1426	1561	1647	1473	1667	1415
2	1437	1466	1556	1397	1522	1622
3	1498	1402	1454	1464	1382	1606
4	1525	1489	1416	1522	1399	1490
5	1569	1539	1383	1600	1485	1323
Delta	144	159	264	203	284	300
Rank	6	5	3	4	2	1

$$E^F = 95901 + 0.79 \times S_W - 5.694 \times A_{INF} - 3379 \times T_L - 13.31 \times P_S - 893.4 \times T_N + 429.8 \times T_B + 0.00354 \times S_W^2 + 0.06218 \times A_{INF}^2 + 5110 \times T_L^2 + 0.2139 \times P_S^2 + 1.767 \times T_N^2 - 2.423 \times T_B^2 \quad (5)$$

Table 4 includes the data utilized for comparing the results obtained by implementing the three different regression models: RQRM, LRM, and QRM. These were taken into consideration to select the most appropriate one for the purposes of this investigation. The RQRM and QRM results for all five (5) response metrics were significantly better than those of the LRM. However, those derived from RQRM were suitable enough to be chosen instead of QRM. As a result, Eqs. (6)–(10) and (11)–(15), which belong to the LRM and QRM, respectively, were not applied in this study.

Linear regression model equations for σ_B^T, σ_Y^T, E^T, σ_B^F, E^F

Table 3 Measured response's mean values and their standard deviations for σ_B^T, σ_B^Y, E^T, σ_B^F and E^F

Run	$\sigma_B^T(MPa)$	$\sigma_Y^T(MPa)$	$E^T(MPa)$	$\sigma_B^F(MPa)$	$E^F(MPa)$
1	48.70 ± 1.37	45.01 ± 1.00	228.10 ± 20.80	67.37 ± 3.10	1731.95 ± 81.01
2	45.81 ± 2.12	43.73 ± 2.24	213.92 ± 9.65	58.70 ± 1.15	1533.23 ± 17.37
3	45.64 ± 0.71	43.93 ± 0.70	221.14 ± 9.84	48.06 ± 0.79	1277.11 ± 21.25
4	45.30 ± 0.51	43.86 ± 0.48	218.61 ± 2.34	48.20 ± 3.81	1286.28 ± 106.20
5	43.45 ± 0.17	41.71 ± 0.19	202.19 ± 4.71	44.90 ± 1.32	1299.76 ± 31.57
6	49.92 ± 0.89	47.06 ± 1.17	223.76 ± 7.11	50.42 ± 1.00	1284.37 ± 33.33
7	47.40 ± 0.57	45.30 ± 0.50	227.33 ± 5.30	49.98 ± 0.47	1323.13 ± 14.86
8	46.42 ± 0.61	44.07 ± 0.56	233.87 ± 3.40	64.02 ± 4.27	1688.93 ± 122.21
9	42.25 ± 0.36	41.08 ± 0.34	201.12 ± 2.16	55.63 ± 0.67	1452.77 ± 18.59
10	38.59 ± 2.98	37.17 ± 3.02	161.32 ± 23.28	47.07 ± 1.56	1435.72 ± 27.94
11	46.99 ± 0.78	43.36 ± 1.52	225.07 ± 8.01	64.22 ± 1.19	1669.27 ± 30.36
12	44.33 ± 0.35	42.64 ± 0.23	185.20 ± 12.67	43.25 ± 3.63	1103.03 ± 95.16
13	43.12 ± 0.61	41.73 ± 0.48	190.48 ± 2.09	39.28 ± 1.08	1038.25 ± 24.38
14	45.79 ± 0.65	44.00 ± 0.76	201.73 ± 12.59	66.60 ± 0.94	1749.54 ± 25.03
15	40.81 ± 3.01	38.51 ± 2.87	178.38 ± 11.67	59.09 ± 6.42	1932.03 ± 48.81
16	45.64 ± 0.54	42.56 ± 1.07	213.03 ± 3.65	58.87 ± 1.35	1534.47 ± 42.75
17	47.18 ± 0.89	43.33 ± 2.35	200.99 ± 21.70	62.10 ± 2.67	1538.99 ± 70.00
18	46.86 ± 0.61	44.53 ± 1.22	174.80 ± 8.84	55.67 ± 3.16	1484.59 ± 85.48
19	38.88 ± 0.38	37.31 ± 0.62	166.18 ± 9.36	45.23 ± 4.97	1512.04 ± 96.82
20	29.27 ± 0.36	28.07 ± 0.35	105.11 ± 4.69	37.22 ± 3.82	1556.03 ± 68.04
21	41.89 ± 0.53	39.21 ± 0.51	202.52 ± 2.02	58.23 ± 2.37	1584.05 ± 51.34
22	43.73 ± 1.18	41.48 ± 1.46	214.09 ± 7.50	70.14 ± 1.86	1831.17 ± 30.34
23	42.63 ± 1.03	40.27 ± 1.01	190.01 ± 4.91	52.98 ± 2.93	1519.97 ± 86.19
24	39.85 ± 1.60	36.87 ± 1.71	142.52 ± 14.11	38.44 ± 3.65	1442.27 ± 109.24
25	33.40 ± 0.51	31.26 ± 0.58	111.05 ± 8.24	26.68 ± 3.26	1469.24 ± 95.29

Table 4 Comparison between LRM, RQRM, and QRM

	LRM			RQRM			QRM		
	R^2	R^2 (adj)	F-Value	R^2	R^2 (adj)	F-Value	R^2	R^2 (adj)	F-value
$\sigma_B^T(MPa)$	73.08%	71.71%	53.38	91.56%	90.66%	101.25	92.72%	91.57%	80.19
$\sigma_Y^T(MPa)$	69.28%	67.71%	44.34	89.06%	87.88%	75.95	90.67%	89.19%	61.16
$E^T(MPa)$	74.71%	73.43%	58.11	88.44%	87.20%	71.41	90.30%	88.76%	58.60
$\sigma_B^F(MPa)$	45.05%	42.25%	16.12	92.06%	91.21%	108.20	92.58%	91.41%	78.59
$E^F(MPa)$	45.52%	42.75%	16.43	88.72%	87.52%	73.44	89.69%	88.05%	54.75

$$\sigma_B^T = 38.60 - 0.05725 \times S_W - 0.09919 \times A_{INF} - 5.81 \times T_L + 0.0676 \times P_S - 0.0001 \times T_N + 0.1493 \times T_B \quad (6)$$

$$\sigma_Y^T = 27.32 - 0.06173 \times S_W - 0.08383 \times A_{INF} - 5.10 \times T_L + 0.0622 \times P_S + 0.0412 \times T_N + 0.1330 \times T_B \quad (7)$$

$$E^T = 101.1 - 0.5079 \times S_W - 0.6935 \times A_{INF} + 9.6 \times T_L + 0.688 \times P_S + 0.508 \times T_N + 0.198 \times T_B \quad (8)$$

$$\sigma_B^F = 118.2 - 0.0396 \times S_W - 0.1763 \times A_{INF} - 47.5 \times T_L + 0.1857 \times P_S - 0.204 \times T_N - 0.008 \times T_B \quad (9)$$

$$E^F = 4458 + 1.502 \times S_W - 0.097 \times A_{INF} - 1335 \times T_L + 3.80 \times P_S - 9.71 \times T_N - 6.33 \times T_B \quad (10)$$

Quadratic regression model equation for σ_B^T, σ_Y^T, E^T, σ_B^F, E^F

$$\sigma_B^T = 3251 - 2.029 \times S_W + 0.0646 \times A_{INF} - 30.3 \times T_L + 0.340 \times P_S - 24.14 \times T_N - 2.88 \times T_B - 0.000286 \times S_W^2 - 0.001773 \times A_{INF}^2 + 52.2 \times T_L^2 - 0.00649 \times P_S^2 + 0.0471 \times T_N^2 + 0.0181 \times T_B^2 + 0.000556 \times S_W \times A_{INF} + 0.243 \times S_W \times T_L + 0.00339 \times S_W \times P_S + 0.00727 \times S_W \times T_N - 0.000348 \times A_{INF} \times P_S \quad (11)$$

$$\sigma_Y^T = 3276 - 2.529 \times S_W + 0.1532 \times A_{INF} + 3.2 \times T_L + 0.324 \times P_S - 23.68 \times T_N - 4.49 \times T_B - 0.000315 \times S_W^2 - 0.002016 \times A_{INF}^2 - 22.8 \times T_L^2 - 0.00728 \times P_S^2 + 0.0459 \times T_N^2 + 0.0272 \times T_B^2 + 0.000305 \times S_W \times A_{INF} + 0.249 \times S_W \times T_L + 0.00435 \times S_W \times P_S + 0.00916 \times S_W \times T_N - 0.000740 \times A_{INF} \times P_S \quad (12)$$

$$E^T = 36829 - 31.57 \times S_W + 2.132 \times A_{INF} + 587 \times T_L - 0.55 \times P_S - 273.4 \times T_N - 32.9 \times T_B + 0.00190 \times S_W^2 - 0.01406 \times A_{INF}^2 - 1945 \times T_L^2 - 0.0305 \times P_S^2 + 0.5297 \times T_N^2 + 0.2023 \times T_B^2 - 0.00056 \times S_W \times A_{INF} + 4.73 \times S_W \times T_L + 0.0551 \times S_W \times P_S + 0.1099 \times S_W \times T_N - 0.01903 \times A_{INF} \times P_S \quad (13)$$

$$\sigma_B^F = 3793 + 1.13 \times S_W - 0.216 \times A_{INF} - 476.9 \times T_L - 1.404 \times P_S - 40.3 \times T_N + 29.27 \times T_B - 0.001130 \times S_W^2 - 0.001778 \times A_{INF}^2 + 932 \times T_L^2 + 0.01734 \times P_S^2 + 0.0815 \times T_N^2 - 0.1619 \times T_B^2 + 0.001455 \times S_W \times A_{INF} + 0.633 \times S_W \times T_L + 0.00176 \times S_W \times P_S - 0.00479 \times S_W \times T_N + 0.00170 \times A_{INF} \times P_S \quad (14)$$

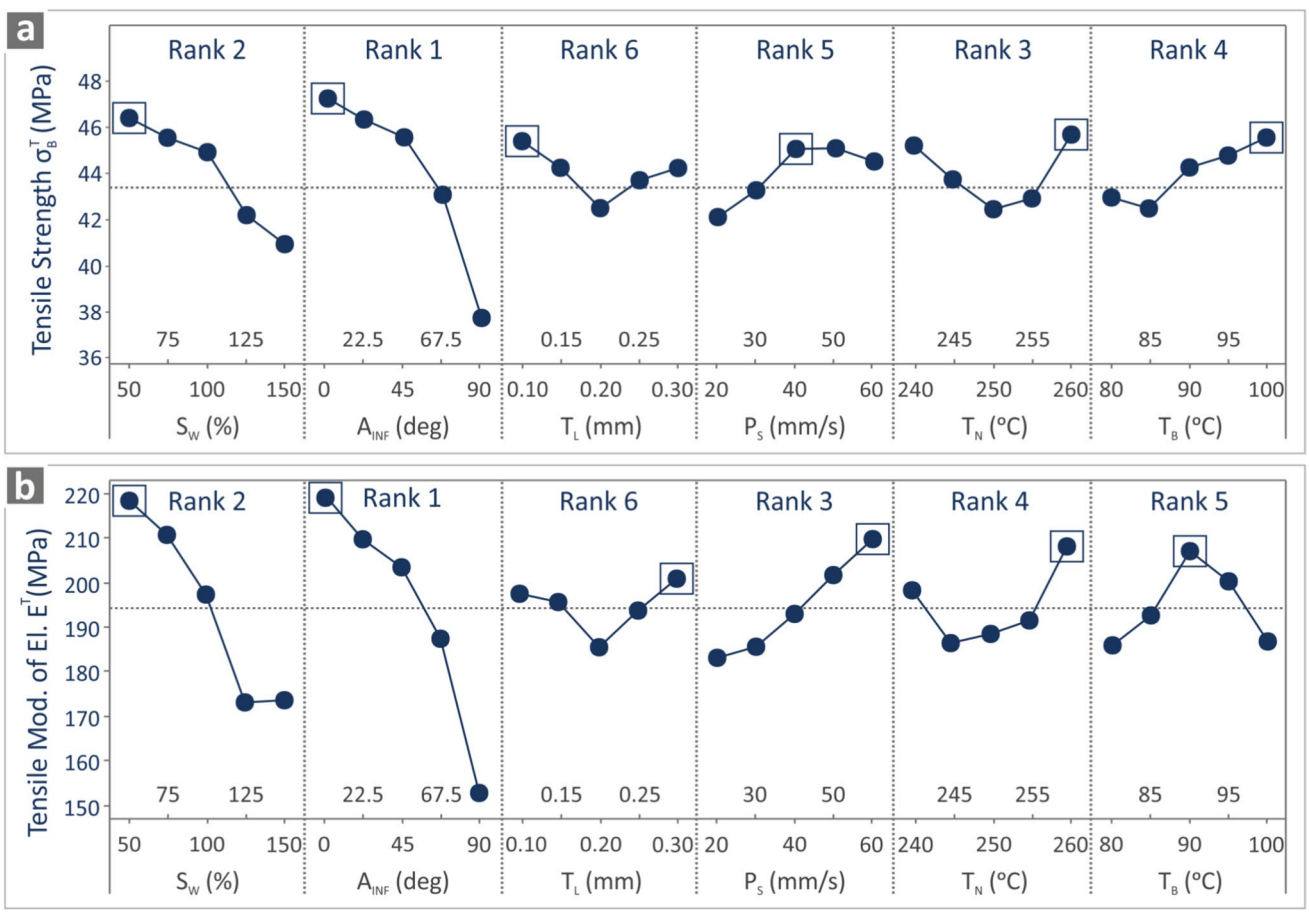


Fig. 7 Main effect plots showing (a) tensile strength vs the S_W, A_{INF}, T_L, P_S, T_N, and T_B control parameters and their ranks, (b) tensile modulus of elasticity vs the S_W, A_{INF}, T_L, P_S, T_N, and T_B control parameters and their ranks

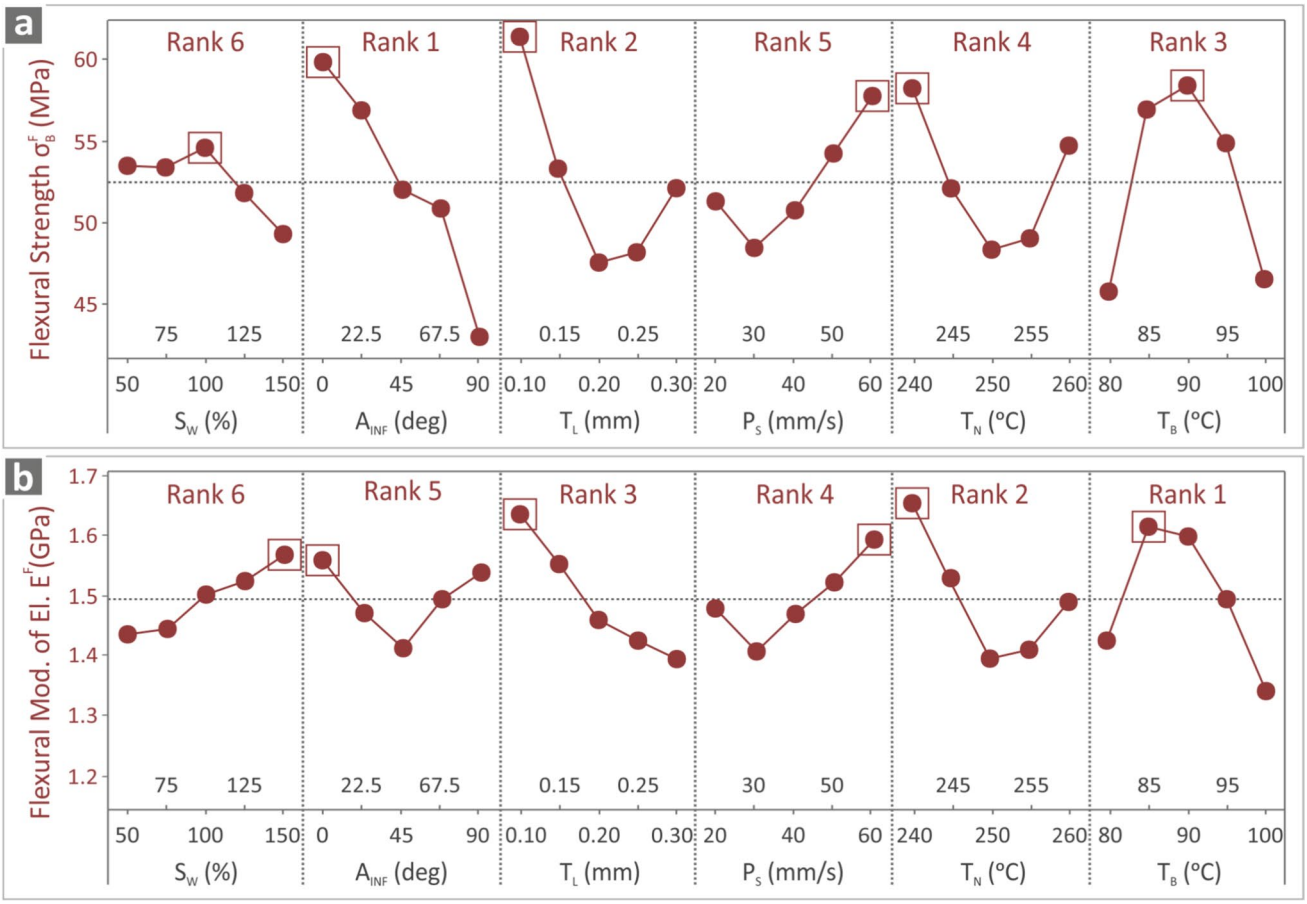


Fig. 8 Main effect plots showing (a) flexural strength vs the S_W, A_{INF}, T_L, P_S, T_N, and T_B control parameters and their ranks, (b) flexural modulus of elasticity vs the S_W, A_{INF}, T_L, P_S, T_N, and T_B control parameters and their ranks

Table 5 Control parameters for the confirmation runs

Run	$S_W(\%)$	$A_{INF}(deg)$	$T_L(mm)$	$P_S(mm/sec)$	$T_N(°C)$	$T_B(°C)$
26	84	31	0.14	30	246	86
27	125	75	0.23	49	256	94

Table 6 Measured responses mean values and their standard deviations values for σ_B^T, σ_B^Y, E^T, σ_B^F and E^F, for the confirmation runs

Run	$\sigma_B^T(MPa)$	$\sigma_Y^T(MPa)$	$E^T(MPa)$	$\sigma_B^F(MPa)$	$E^F(MPa)$
26	48.07 ± 1.58	37.72 ± 1.07	224.15 ± 6.26	63.59 ± 2.00	1673.68 ± 49.76
27	41.61 ± 1.22	39.27 ± 1.06	186.35 ± 6.28	53.05 ± 1.63	1685.76 ± 53.55

$$\begin{aligned} E^F &= 73767 + 29.3 \times S_W - 11.71 \times A_{INF} - 5994 \\ &\times T_L - 26.78 \times P_S - 799 \times T_N + 628 \times T_B \\ &+ 0.00040 \times S_W^2 + 0.0511 \times A_{INF}^2 + 10367 \\ &\times T_L^2 + 0.3576 \times P_S^2 + 1.612 \times T_N^2 - 3.510 \\ &\times T_B^2 + 0.0529 \times S_W \times A_{INF} + 6.8 \times S_W \times T_L \\ &+ 0.0126 \times S_W \times P_S - 0.128 \times S_W \times T_N \\ &+ 0.0407 \times A_{INF} \times P_S \end{aligned} \tag{15}$$

Figure 7 and Fig. 8 sum up the influence of the control factors on the output metrics, through main effect plots. Figure 7a presets the data with regard to the tensile strength versus the chosen control parameters S_W, A_{INF}, T_L, P_S, T_N, and T_B. It is evident that A_{INF} is the parameter causing the greatest influence when changing, which is why it is categorized in the first rank. Then the parameter placed in the sixth rank is T_L for the tensile strength, indicating the least influence and changes on the behavior of the response metric.

Table 7 Validation table

Run		26	27
Experimental	$\sigma_B^T(MPa)$	48.07	41.61
	$\sigma_Y^T(MPa)$	37.72	39.27
	$E^T(MPa)$	224.15	186.35
	$\sigma_B^F(MPa)$	63.59	53.05
	$E^F(MPa)$	1673.68	1685.76
Predicted	$\sigma_B^T(MPa)$	45.13	39.38
	$\sigma_Y^T(MPa)$	43.13	37.73
	$E^T(MPa)$	202.67	166.70
	$\sigma_B^F(MPa)$	59.61	45.55
	$E^F(MPa)$	1493.55	1415.59
Absolute error	$\sigma_B^T(\%)$	6.11	5.36
	$\sigma_Y^T(\%)$	14.35	3.94
	$E^T(\%)$	9.58	10.55
	$\sigma_B^F(\%)$	6.26	14.14
	$E^F(\%)$	10.76	16.03

Figure 7b is the respective information with regard to the tensile modulus versus the chosen control parameters S_W, A_{INF}, T_L, P_S, T_N, and T_B. A_{INF} has the greatest impact on the behavior of tensile modulus, being in the first rank. The sixth rank is the T_L parameter. For the flexural strength (Fig. 8a) A_{INF} is again the parameter causing the greatest influence when changing, which is why it is categorized in the first rank. Then the parameter placed in the sixth rank is S_W for the flexural strength, indicating the least influence and changes on the behavior of the response metric. Figure 8b is the respective information with regard to the flexural modulus versus the chosen control parameters S_W, A_{INF}, T_L, P_S, T_N, and T_B. T_B has the greatest impact on the behavior of flexural modulus, being in the first rank. In the sixth rank is the S_W for the flexural modulus. Additional main effect plots are provided in the Supplementary information.

3.4 Confirmation experiment and validation.

Two confirmation experiments were performed, except for the other twenty-five (25) runs, with the aim of validating the prediction functions. Table 5 and Table 6 include the applied 3D printing parameter levels for Runs 26 and 27 as well as the average and standard deviation values, respectively. In the supplementary information, Table S2 lists the experimental results analytically. Table 7 presents the validation results by showing the values derived experimentally and the predicted ones of the two runs, as well as their calculated absolute error, after finding the difference between the measured and calculated values. As can be observed, the measured error remained low for all five response metrics.

In Fig. 9, there are four graphs with regard to the four response metrics of tensile (Fig. 9a) and flexural (Fig. 9b) strength, as well as the tensile (Fig. 9c) and flexural (Fig. 9d) modulus of elasticity. The results of all (twenty-five) 25 runs, as well as the confirmation runs 26 and 27. As can be observed, it was ascertained that the predicted values and the ones experimentally found were close, for all of the response metrics. The same result was

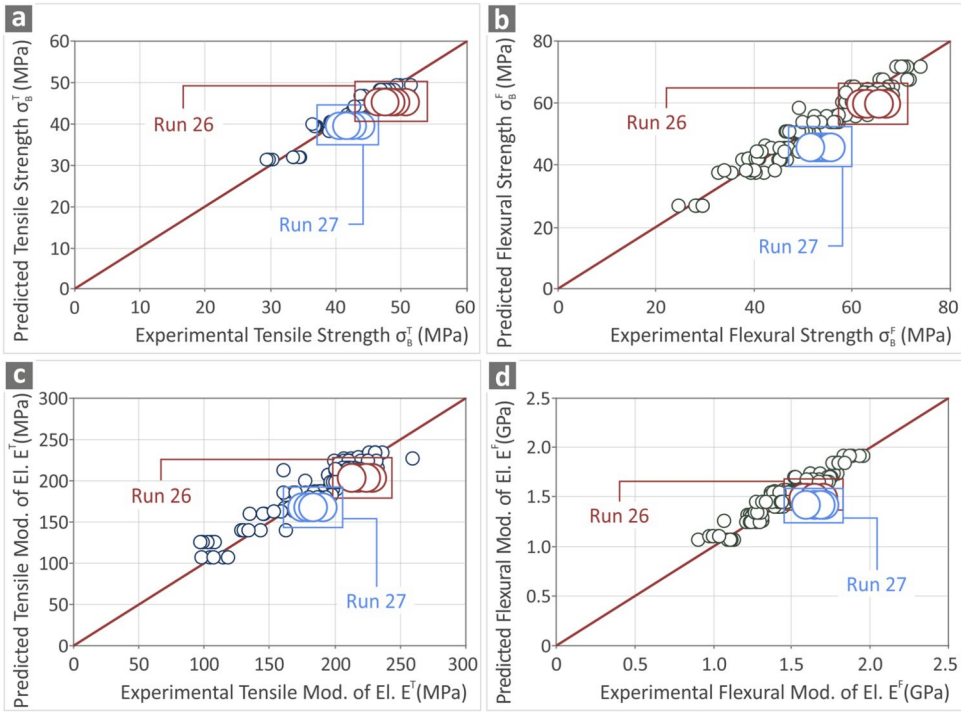


Fig. 9 Graphs presenting the relation between the experimental and the predicted values for the response metrics of (a) tensile strength, (b) flexural strength, (c) tensile modulus of elasticity, and (d) flexural modulus of elasticity (with regard to the runs 1–25 and the confirmation runs 26–27)

obtained in the confirmation runs. In the supplementary information, there are also box plots (presenting mainly scattered data, while only in few values are concentrated around a value, Fig. S1) and surface plots (3D graphs showing simultaneously how the two most critical control parameters for each response metric affect its variation, Fig. S2). The respective information and data with regard to the tensile yield strength are also presented in the supplementary information (Fig. S3).

4 Discussion

The utilization of ASA material in this work aims to determine its optimum 3D printing parameters for manufacturing specimens with improved mechanical properties. The Taguchi DOE L25 was used for this purpose, with the control parameters S_W, A_{INF}, T_L, P_S, $T_{N,}$ and $T_{B,}$ and the response metrics σ_B^T, σ_B^Y, E^T, σ_B^F and E^F. To the best of our knowledge, this kind of research is limited to the available literature. Consequently, this issue needs to be given more attention because the results derived could be useful for various applications in different fields. It should be noted that parameters, such as infill density, were not included, as reducing the infill density is known to negatively affect the mechanical performance in 3D printed parts. The cooling rate can be an important parameter in 3D printed polymeric parts, still in literature it is usually investigated in high-performance polymers, in which the temperatures during the 3D printing process are higher and cooling has a high effect on their crystallinity [125, 126]. The cooling rate can affect the warping of the samples. Furthermore, if layers cool too fast before the next layer is deposited, bonding between layers weakens, reducing overall part strength. It also affects the surface quality, due to uneven shrinkage. Herein, no significant warpage or shrinkage phenomena were observed, and the part strength was measured and evaluated, considering two temperature parameters, i.e., nozzle and bed temperature. Still, this is a valuable comment, and the cooling rate parameter can be included in future research.

Evaluating the void percentage in 3D printed parts from the SEM images cannot lead to reliable estimations. The 3D printing structure can have a non-uniform structure and the findings in one cross-section captured during SEM can be misleading. Moreover, the images of the fractured surfaces can be further misleading, due to the fact that the part is deformed. Overall, SEM is limited in its ability to fully capture the depth, volume, and interconnectivity of pores, which can result in the underestimation or mischaracterization of porosity. This limitation poses challenges in obtaining statistically representative porosity data, particularly in materials where pores are heterogeneously distributed across larger volumes, such as the 3D-printed parts. The process of fracturing a sample, especially if conducted in an uncontrolled manner, may introduce artifacts such as smearing or tearing, which can obscure or distort the actual pores. These artifacts may be mistaken for or conceal true porosity features. As a surface-sensitive technique, SEM is unable to detect subsurface pores unless they are exposed by fracturing. Even when pores are visible, quantifying their size, shape, and distribution from SEM images necessitates complex image processing, which can be prone to errors and subjectivity in the absence of standardized procedures. Void fraction and other metrics would require more sophisticated tools, such as micro-computed tomography. Further analysis of the porosity of the samples was not within the scope of the current research, which focused on mechanical performance. This can be a response metric in future research. From the lateral side, conclusions can be made regarding the uniformity of the layer thickness and the adhesion between the layers by visually inspecting the images. In all cases, a uniform layer formation was observed with good adhesion between the layers, as shown in Fig. 5.

The ductile/brittle behavior was evaluated through the differences in the strain at failure of the samples, as is depicted in the stress vs. strain graphs (Fig. 4). Furthermore, the SEM images of the fracture regions were examined and the differences in the deformation of the fracture behavior were evaluated. The intrinsic characteristics of polymers, including molecular weight, crystallinity, and polymer chain mobility, are pivotal in determining their ductile/brittle behavior. Amorphous polymers, such as ASA, typically exhibit greater ductility, whereas semicrystalline polymers, such as PLA, are generally more brittle due to their elevated crystallinity. Still, the 3D printing parameters are important in the ductile/brittle behavior of the samples as well. The temperatures of the extruder and bed are critical in influencing interlayer adhesion. A low nozzle temperature may result in weak interlayer bonding, leading to brittle failure modes. In contrast, optimal temperatures facilitate diffusion across layers, promoting ductile fracturing behavior through enhanced polymer chain entanglements. Reduced layer heights generally enhance mechanical properties by increasing the bonding surface area and minimizing porosity. Furthermore, the build direction affects anisotropy; materials constructed with directions parallel to the loading during tension demonstrate higher ductility compared to those built perpendicular to the load, which tend to fail in a brittle manner across layer interfaces. The infill pattern (not evaluated herein) also influences mechanical performance, with geometries that enable stress redistribution and crack-stopping capabilities favoring mechanisms that allow for ductile deformation. Elevated printing speeds can lead to inadequate bonding due to reduced thermal interaction time, potentially increasing brittleness. Studying the effect of the ASA intrinsic characteristics on the ductile/brittle behavior of the 3D printed samples, was not within the scope of the current research.

The selection of the DOE was based on the desire to test significantly fewer specimens while still extracting the desired data and maintaining reliability. In particular, twenty-five (25) runs were conducted, with five (5) replicas each, constituting a total of one hundred and twenty-five (125) tested specimens. However, the selection of the full factorial design requires an extremely large number of specimens to be produced and tested.

The following calculation steps would be to raise the level number in the number of control parameters ($5^6 = 15{,}625$) and then multiply the result with the repetitions ($15{,}625 \times 5 = 78{,}125$). Consequently, 78,125 specimens were produced for each tensile and flexural test. In addition, the specimens of the confirmation runs ($6^1 \times 5 = 30$) plus 30 more specimens for each of the tensile and flexural tests yielded a total of 156,310 tested specimens. Herein, even with the method followed which requires fewer samples to be tested, a laborious effort with two hundred ninety (290) samples were tested. One hundred and twenty-five for each mechanical test (tensile and flexural) and forty (40) more for the confirmation runs.

The most influential parameter was found to be A_{INF} for modulus of elasticity, tensile strength, and flexural strength, whereas T_B was the most influential control parameter for the flexural modulus of elasticity. The printing parameters that did not have a significant effect on the mechanical behavior of the manufactured examples were S_W for the flexural properties and T_L for the tensile properties (both strength and modulus of elasticity).

Run 6 had the highest tensile yield (47.06 Mpa) and ultimate strength (49.92 Mpa) with printing settings S_W 75%, A_{INF} 0 deg, T_L 0.15 mm, P_S 40 mm/s, T_N 255 °C and T_B 100 °C. At the same time, it had a high Young modulus, median flexural strength, and low flexural modulus of elasticity. The highest Young modulus (233.87 MPa) was found in Run 8 with printing settings S_W 75%, A_{INF} 45 deg, T_L 0.25 mm, P_S 60 mm/s, T_N 240 °C and T_B 85 °C. The lowest tensile yield (28.07Mpa), ultimate strength (29.27 MPa), and Young modulus (105.11 MPa) were found in Run 20, with printing settings S_W 125%, A_{INF} 90 deg, T_L 0.20 mm, P_S 20 mm/s, T_N 255 °C and T_B 85 °C. The highest flexural strength (70.14 MPa) was found in Run 22, with printing settings S_W 150%, A_{INF} 22.5 deg, T_L 0.10 mm, P_S 60 mm/s, T_N 255 °C and T_B 90 °C. The lowest one (26.68 MPa) was found in Run 25, with printing settings S_W 150%, A_{INF} 90 deg, T_L 0.25 mm, P_S 40 mm/s, T_N 245 °C and T_B 80 °C. Finally, the highest flexural modulus of elasticity (1932.03 MPa) was found in Run 15, with printing settings S_W 100%, A_{INF} 90.0 deg, T_L 0.15 mm, P_S 50 mm/s, T_N 240 °C and T_B 90 °C. The lowest one was found in Run 13 (1038.25 MPa), with printing settings S_W 100%, A_{INF} 45 deg, T_L 0.30 mm, P_S 30 mm/s, T_N 255 °C and T_B 80 °C. According to these findings, the difference between the highest and lowest tensile strength reported was 41.37%, for the yield strength 40.35%, for the Young modulus 55.06%, for the flexural strength 61.96%, and for the flexural modulus of elasticity 46.26%. Such differences justify the need for such an analysis, although no single Run optimized all the response metrics. By examining the experimental results, the optimum run can be considered Run 1 with printing settings S_W 50%, A_{INF} 0 deg, T_L 0.10 mm, P_S 20 mm/s, T_N 240 °C, and T_B 80 °C. This Run had high values in all the response metrics. At the same time, it should be noted that samples from Run 1 failed at significantly higher stain values, showing a more ductile behavior, which is also an important feature in applications. It should be noted also that by slightly changing the 3D printing parameter values, the response of the examples changed from ductile to brittle. For example, Run 1 and Run 7 showed similar strength values, still, Run 7 samples failed in the tensile test by almost one-third of the strain, and the Run 1 samples failed. This underscores the significance of choosing appropriate 3D printing parameters when fabricating components using the ASA polymer through the MEX technique.

The RQRM was chosen for the experimental data processing, and by considering the validation table (Table 7) provided above, the comparison of the experimentally determined and predicted values for the two confirmation runs indicated mostly negligible deviations, indicating that the data acquisition was trustworthy. The R values of RQRM differ from the respective R values of QRM by only 1–2%, which is well within the statistical error. They were about 90% for all metrics, which is an excellent value, showing the accuracy of the prediction method. LRM R values were as low as 42% and up to 72%, which are marginally acceptable results, therefore it can be considered that LRM failed to predict the response metrics in this case. The confirmation runs proven prediction models (with the RQRM) accuracy. The prediction accuracy was between 4 and 6% in four out of the ten comparisons, which is an excellent outcome, in three more cases it was about 10% and in the remaining three cases, the absolute variation between the experimental and the predicted values was between 14 and 16%, which is also a very good result, showing the overall prediction accuracy. Nevertheless, it is important to recognize that the forecasting models are anticipated to yield precise outcomes within the specified range of control parameter values. Outside the investigated control parameter levels, the accuracy of the prediction is not a given.

Regression findings show which are significant factors for the response metrics, by evaluating the *F* and *P* values. By examining these values, control parameters, such as the strand width are insignificant for the mechanical properties. Others have mild significance, such as the printing orientation and speed. What is interesting is that some control parameters have contradictive significance. The layer

thickness is insignificant for the tensile properties but critical for the flexural properties. Same for the nozzle and the bed temperature. This can be attributed to the impact on each parameter in correlation to the loading-bearing mechanism in the 3D-printed parts. In the context of MEX, the impact of layer thickness on the mechanical performance of printed polymer components is significantly influenced by the mode of mechanical loading. Numerous studies have demonstrated that while layer thickness exerts a relatively minor influence on tensile properties, it is a critical parameter in predicting flexural behavior. Under tensile loading, stress is uniformly distributed across the cross-section of the specimen. Consequently, tensile strength is predominantly determined by the material's intrinsic properties and the degree of interlayer adhesion [127]. Within typical processing ranges, generally from 0.1 mm to 0.3 mm, variations in layer thickness do not substantially alter the load-bearing cross-sectional area or introduce stress concentrations. Therefore, the tensile response remains largely unaffected by changes in layer thickness, provided that a high degree of interlayer adhesion is achieved during deposition [128]. However, under flexural loading conditions, the stress distribution is non-uniform, with peak tensile and compressive stresses occurring at the outermost layers of the component. This makes flexural properties highly sensitive to surface quality, layer interface defects, and geometric discontinuities, such as the "stair-step" effect associated with increased layer heights [129]. Moreover, increased layer thickness can result in reduced interlayer fusion due to prolonged cooling times between deposition passes, thereby compromising anisotropy and interfacial strength [130]. These cumulative effects lead to reductions in flexural properties and strength with increasing layer thickness. While the material and consistency of bonding predominantly govern tensile properties, flexural properties are strongly influenced by the geometric resolution and structural integrity of printed layers. In applications where bending loads are significant, careful optimization of layer thickness is essential to ensure mechanical reliability.

The temperature of the nozzle is a critical factor in determining the quality of interlayer bonding and the structural integrity of parts. However, its influence is not uniformly distributed across all mechanical loading modes, the same as the layer thickness. Empirical evidence and theoretical insights suggest that while nozzle temperature has a relatively minor impact on tensile properties, it significantly affects the flexural properties of printed parts, due to the nature of the loadings, explained above. As long as the nozzle temperature is sufficiently high to facilitate adequate polymer flow and interdiffusion at the layer interfaces, minor variations in nozzle temperature typically do not result in substantial changes in tensile strength or modulus. Once the polymer achieves a certain degree of molecular mobility, allowing for entanglement between layers, further increases in nozzle temperature yield diminishing returns in terms of interlayer adhesion [128, 130]. Consequently, tensile properties reach a plateau and become insensitive to further increases in nozzle temperature. In contrast, flexural loading imposes a non-uniform stress profile, with the surface layers experiencing the highest tensile and compressive stresses. The mechanical integrity under flexural loading is thus more susceptible to surface defects or weaknesses, which are directly influenced by nozzle temperature during deposition. Inappropriate nozzle temperatures can lead to inadequate layer adhesion, void formation, and improper polymer wetting, particularly along the surface contours of the printed geometry [129].

Regarding the bed temperature, a similar effect with the nozzle temperature occurs, but it is related primarily to enhancing dimensional stability and adhesion of the initial layers during printing. As mentioned, flexural properties are more susceptible to local defects and geometric imperfections, particularly at the surface and bottom regions of the part, where bending stresses are maximal. Suboptimal bed temperatures can lead to poor adhesion in the initial layers, residual stress accumulation, and temperature gradients that may cause warping, microvoids, or interfacial defects [129, 130]. These defects, typically localized in the bottom region of the print, do not necessarily affect performance under axial tensile loading but can act as stress concentrators under bending conditions, where the outermost and bottom fibers experience maximum tensile and compressive stresses. Thus, inadequate bed temperature control can result in premature failure or reduced flexural strength, although the overall tensile behavior remains unaffected. Therefore, while the influence of bed temperature on interlayer bonding—and consequently on tensile performance—is limited to the early stages of printing, its significance becomes apparent during the evaluation of flexural behavior, where structural integrity near the bottom of the part is crucial for mechanical performance. This distinction highlights the load-specific importance of thermal process parameters in FFF-printed polymer parts.

Comparing this research with relevant publications in the literature, Magri et al. [51] investigate only three parameters, while herein six parameters were studied. The findings reveal the synergistic relation between the parameters, therefore justifying the need for the extra parameters study. Furthermore, a different modeling approach was followed, which had a great effect on the modeling results. Hameed et al. [99] studied five parameters, but they were not the same as the current research. For example, infill density is investigated, which is not included in the current research. Infill density was not investigated herein, as it has been reported several times before that infill density lower than 100% negatively affects the mechanical performance of the 3D printed

samples. This is the expected and logical outcome, which was also verified in the research of Hameed et al. Altering the studied parameters reveals synergistic relations between them, which cannot be located otherwise. Even the investigation of the control parameters in a different level range has value, as the findings could differ, as the existing literature has shown. Furthermore, herein, three regression modeling tools were applied, which led to the compilation of prediction models, verified with additional confirmation runs, and proven their reliability. The comparison by itself has merit, as the efficacy of three modeling approaches was revealed and their performance was evaluated for this specific experimental scenario.

The tensile strength is compared with respective values reported in the literature for bulk [131, 132] and 3D-printed ASA [44, 133] parts. For the bulk material, the tensile strength is reported to be about 25 MPa [131] or 45 MPa [132] in another research, while for the 3D printed parts the reported values can reach 52.2 MPa after optimization of the 3D printed parameters [44], or 40 MPa [133] according to another publication, without optimization of the parameters. Herein, the optimized tensile strength was 49.92 MPa, which is in good agreement with the literature. For the flexural strength, literature on 3D printed values reports 65.7 MPa [133] after optimization, while herein 70.14 MPa was achieved. For the bulk material [131] a flexural strength of about 40 MPa is reported. According to the material's datasheet, the tensile strength is 50MPa. Differences are attributed to the different standards followed in the tests, the different grades in the reported research, the different experimental conditions, and the different 3D printed parameters utilized for the 3D printing of the parts. Such findings show the importance of optimization on the 3D printed parameters, as the strength can reach or even exceed the reported values for the bulk material in this specific polymer. These are now discussed in the revised manuscript.

5 Conclusions

In this study, the Taguchi L25 DOE was employed to investigate more effectively and promptly the influence of S_W, A_{INF}, T_L, P_S, T_N, and T_B printing parameters on the characteristics of ASA 3D printed parts, with the aim of producing more robust parts, through the optimization of their mechanical properties. The chosen responses were namely σ_B^T, σ_B^Y, E^T, σ_B^F and E^F. The mechanical characterization through tensile and flexural testing revealed the overall optimum results in the case of Run 1 (all five metrics simultaneously with high values, close to the highest reported in the research), which was created by the 3D printing parameters of S_W 50%, A_{INF} 0 deg, T_L 0.10 mm, P_S 20 mm/s, T_N 240 °C and T_B 80 °C.

With regard to the morphology of the fabricated samples, some were revealed to have ductile characteristics (for example, Run 1) and other brittle behaviors (for example, Run 7), highlighting the importance of the printing settings on the response of the items. Some printing parameters were shown to have a more notable impact on the characteristics of the examples investigated. In particular, A_{INF} and T_B found in the first rank caused the greatest changes in the performance of the parts when they had different values. On the other hand, the printing parameters with the lowest impact were T_L and S_W, which were ranked sixth.

Overall, the impact of the 3D printing settings on the performance of the samples was noticeable and created hope for the application of ASA materials in many potential applications that several fields might be in need of. The analysis revealed that by choosing proper 3D printing parameters, the strength of the items can be enhanced by more than 40% in uniaxial loading scenarios and more than 60% in bending loading scenarios. Similar results were found for the stiffness of the parts. Such findings justify the necessity for such an optimization. Furthermore, RQRM was found to be sufficient enough, while being simpler than QRM at the same time, among the three regression models evaluated. The prediction functions formed proved their accuracy in the confirmation runs and can be directly applied to real-life applications. Future work could include an examination of more printing parameters at different levels and combinations, and response metrics, such as the porosity in the parts. Furthermore, the effect of the ASA intrinsic characteristics on the ductile/brittle behavior of the 3D-printed samples can be investigated. Post-processing can also be examined, to improve the characteristics of the samples, such as their surface quality due to the 3D printing structure. This is important as the findings can be considered reliable within the range of levels considered in the research, which can be broadened through further investigations.

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Data availability The raw/processed data required to reproduce these findings cannot be shared because of technical and time limitations

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

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